## PARTITIONS & FILESYSTEMS:

**Partition:**

Partition is dividing One HDD in to multiple logical Drives. The purpose we are going for partitioning is for easier management of data and to utilise the disk space more effectively

**Physical Drives Representation in Linux :**

SATA / SCSI (/dev/sdX)
IDE (/dev/hdX)
Virtual HDD (/dev/vdX)

**Logical drives representation in LINUX :**

**Server1**
/dev/sda
/dev/sda1
/dev/sda2
/dev/sda3
/dev/sda4

/dev/sdb

/dev/sdb1
/dev/sdb2
/dev/sdb3
/dev/sdb4

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**Server2**
**IDE**

/dev/hda
/dev/hdb
/dev/hdc

**Ser ver3**
**Virtual HDD**

/dev/vda
/dev/vdb
/dev/vdc

**Types of partitions:**
1)Primary Partition
2)Extended partition

**Primary Partition :**
These are the partitions where bios can recognise. These are the partitions we can use it for installing OS

**Extended Partition:**
These are the partitions where bios can't recognise. These are the partitions we can't use it for installing OS

**Logical partitions:**
These are the partitions we creating inside of Extended partitions.

**We can create partitions using :**
1)Fdisk
2)Parted
3)gdisk

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**Creating Partitions using FDISK:**

#fdisk -l **(Lists all the available HDD's along with its Partitions)**

#fdisk -l /dev/sda **(Lists information about /dev/sda & its partitions)**

```
#fdisk      /dev/sdb
            :m(for Help)
            :p(prints the Partition table)
            :n(for creating New Partition)
            :Primary / Extended (Select Primary)
            :[1 – 4] Partition Number
            :Enter First Sector [Take default]
            :Enter last Sector[enter  +2G]
            :p
            :w     [Writes the changes to Partition Table]
#partprobe          /dev/sdb  (Inform the changes happed on /dev/sdb to Kernel)

#fdisk -l   /dev/sdb
```

**Deleting Partition using FDISK:**

```
#fdisk      /dev/sdb
            :p
            :d  (To Delete the partition)
            :Enter the Partition ID to delete
            :p
            :w
#partprobe          /dev/sdb
#fdisk      -l      /dev/sdb
```

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**Creating Partitions using PARTED :**

#parted -l **(Lists all the available disks along with its partitions)**

#parted -l <disk> (**Displays information about /dev/sdb & its partitions)**
/dev/sdb

#parted /dev/sdb
:help (Lists all the options we can use under parted prompt)
:print
:mkpart [To create new partition]
:Filesystem type [ Just enter]
:primary /Secondary [Select Primary]
:start Location [ Enter 1]
:End Location [6G]
:print
:quit

#partprobe /dev/sdb
#parted -l /dev/sdb

**Deleting the Partition using PARTED :**

#parted /dev/sdb
:print
:rm 1
:Print
:quit
#partprobe /dev/sdb

**Difference between FDISK & PARTED :**

→ FDISK uses MBR(Master boot record) Labelling where it can support disks up to a max of 2TB only.
→PARTD & GDISK uses Guided Partition table (GPT) Labelling where it can support disks of size more than 2TB.

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